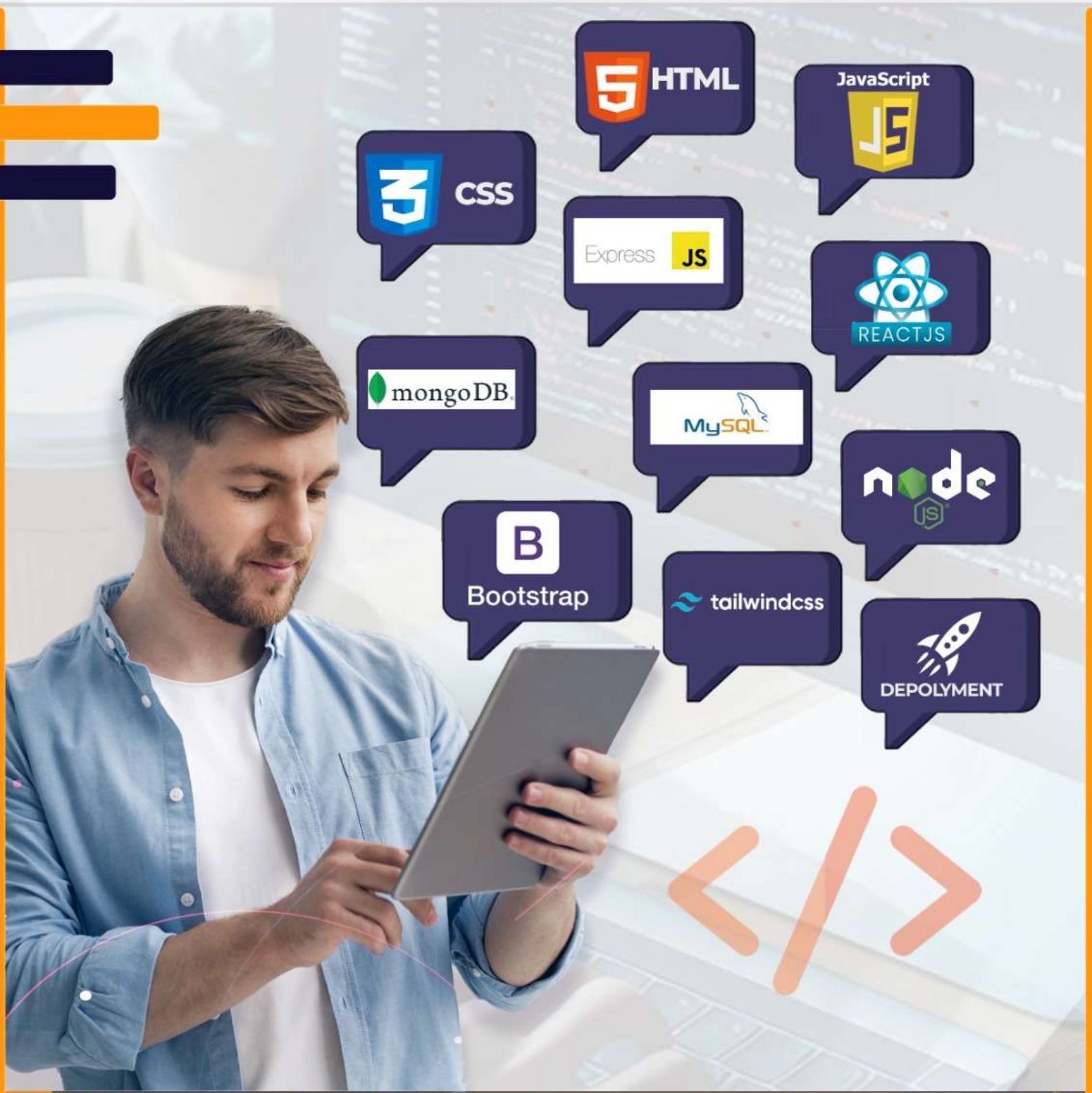
Upflairs
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FULL STACK WEB DEVELOPMENT



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ABOUT US

UpFlairs is an innovative educational technology company with a clear mission to elevate the skills and employability of students throughout India.

Our dedicated team is committed to fostering the next generation of tech talent, equipping them with cutting-edge skills in emerging technologies and has educated more than 47K students all over the globe including IITs, NITS, Deemed Universities and other colleges.

We offer the courses that are most trending technologies of the recent era in the fields like AI/ML, Data Science, Cloud Computing, DevOps, Full Stack Web Development, Embedded Systems, IoT, and Robotics. These courses are meticulously designed to provide students with the practical skills and knowledge required to excel in tech-driven careers, making them not just job-ready but industry leaders of tomorrow.

We are not only limited to training the youth of the country but also provide Lab setups to various Colleges and Universities and Project solutions to other companies for the domains like AI-ML, IoT, ROBOTICS AND CLOUD.

EDUCATION



PROJECT SOLUTION



LAB SETUP



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CAREER OPPORTUNITIES IN FULL STACK WEB DEVELOPMENT

Key Job Roles for Full Stack Developers

- Full Stack Developers
- Frontend Developers
- Backend Developers
- Web Developer
- Software Engineer

Industries Hiring Full Stack Developers

- Tech Companies
- Startups
- E-commerce
- Finance & Banking
- Healthcare

Benefits of Career in Full Stack Development

- High Demand & Job Security
- Competitive Salaries
- Remote Work Opportunities
- Career Growth

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MODULE HIGHLIGHT



COMPREHENSIVE CURRICULUM

Learn front-end and back-end development with in-depth modules on technologies like HTML, CSS, JavaScript, React, Node.js, MongoDB, and more.

INDUSTRY PROJECTS

Gain practical experience by working on real-world projects that mirror industry scenarios.





CAREER COACHING

Access resume-building sessions, interview preparation, and personalized career guidance.

MOCK INTERVIEWS & TEST SERIES

Prepare with regular mock interviews & technical test series to enhance your job readiness.





DAILY ASSIGNMENTS & MENTORSHIP

Strengthen your learning with daily assignments and 1:1 mentorship from industry experts.

Full Stack Web Development

Duration – 6 Months (240 Hours)

Module 1: Introduction to Web Development (10 hours)

HTML 5

- What is a Fullstack Development?
- Understanding the role of front-end and back-end development
- Setting Up Environment
- Working with IDE (i.e. Visual Studio Code)
- Understanding HTML
- HTML-Tag, attributes and elements
- Heading, paragraph and paragraph formatting
- IMG, BR, and other important tags
- Lists (OL, UL, DL)
- HTML Forms
- Table
- Audio and Video Tags
- Difference between block and inline Elements
- Div and span
- Semantic tags in HTML

Mini Project 1:

- ***"Wikipedia"***

Module 2: GitHub (6 Hours)

GitHub

- What is GitHub
- Difference b/w git and GitHub
- Create an account on GitHub
- Create repository
- Import code on GitHub
- Working with git CLI
- Initializing and cloning a git repo
- Push code to GitHub repo

Module 3: Introduction to Design Basics (20 hours)

CSS

- What is CSS
- Type of CSS
- Link CSS in HTML
- CSS Syntax: Rules, Declarations, Properties, and Values
- Selectors (id, classes, and element selector)
- Border and border-radius
- Text formatting
- Fonts and Google Fonts

- Font Awesome Icons
- Background
- Css colour
- Padding and Margin
- Box model
- Overflow and shadow
- Lists in CSS
- Styling Forms
- Layout Design patterns in CSS
- Float
- Flex-blox

Module 4: Introduction to Advanced Designing (20 hours)

CSS ADVANCED

- Advance selectors (Attribute, Grouping, Pseudo-classes, Pseudo-elements)
- Pseudo-classes (:link, :visited, :hover, :active)
- Pseudo elements (::before, ::after)
- CSS positions
- Advanced layout in CSS using Grids
- Gradients
- Filters
- Display
- Z-index
- Transformation
- 2D Transforms: Translate, Rotate, Scale, Skew
- Transition
- CSS animation
- Media query
- CSS Variables (Custom Properties)
- CSS Functions
- calc(), clamp(), min(), max()
- Using var() for Dynamic Values
- Gradient Functions (linear-gradient, radial-gradient, conic-gradient)
- Practice advanced website designs

Mini Project 2:

- ***"Portfolio"***
- ***"A Complete Website using HTML and CSS" Like-"blogging website" , "E-Commerce" , "Travel site" , "Restorant Site"***

Module 5 : Javascript Basics (20 Hours)

Javascript

- Introduction to Javascript
- Syntax and Structure
- implementation of JavaScript in html
- Statements and Declarations
- Variables: var, let, const
- Data Types: Strings, Numbers, Booleans, Null, Undefined, Symbol, BigInt
- Type Coercion and Conversion
- **Operators :** Arithmetic Operators , Assignment Operators , Comparison Operators , Logical Operators, increment and decrement Operator (++,--)
- **Control Flow:** Conditional Statements: if, else if, else, switch , Ternary (Conditional) Operator
- **Loops:** break and continue
- **Functions:** Function Declarations and Expressions , Arrow Functions (=>)
- **Working with Strings :** Creating Accessing Strings , String Methods , Template Strings
- **Working with Arrays :** Creating Accessing and Manipulating Array , Traversing in Array(for-in, for-of), Array Methods

Mini Project 3 :

- ***"Creating Calculator"***
- ***"carousel /slide/footer/header"***

Module 6: Advanced JavaScript (30 hours)

JAVASCRIPT ADVANCE

- Date and time
- Objects: Creating and Accessing Objects
- Object Methods: Object.keys(), Object.values(), Object.entries(), Object.assign()
- The Document Object Model (DOM) : Accessing and Manipulating Dom
- Event Handling : Adding event listener to events , Event capturing and event bubling
- Form Validation
- BOM(browser object Model)
- Advanced JavaScript Concepts : Closures
- Scope and Hoisting : Global vs Local Scope , Function Scope vs Block Scope , Variable and Function Hoisting
- this Keyword : Understanding this in Different Contexts , this in Arrow Functions vs Regular Functions
- Promises and Asynchronous JavaScript: Creating and Using Promises , then(), catch(), finally() Methods,

Chaining Promises , Error Handling in Promises , async and await , Working with Async/Await

- Exception Handling: try, catch, finally, throw
- Fetch API : Fetching Data from APIs , Handling Responses and Errors , Parsing JSON
- Working with JSON
- Object-oriented programming (OOP) in JavaScript: Prototypal Inheritance , Prototype Chain, Object. create() and Object. set prototype of()
- ES6 Classes :
- Class Syntax and Inheritance
- Constructors
- Methods and Static Methods
- Getters and Setters
- Extending Classes (Inheritance)
- super() Keyword
- Encapsulation and Polymorphism
- Private and Public Fields
- Method Overriding
- Polymorphism in JavaScript

Mini Project 4:

- ***"A Task Management System"***
- ***"Complete dynamic website using js "***
- ***"ToDo List"***

Module 7: Bootstrap (20 hours)

- What is bootstrap
- Responsive website
- Framework vs library
- Adding bootstrap
- Container
- Grid system (size)
- Button style
- Cards
- Navbar
- Model
- Form
- Table
- Image
- Pagination

- Progress bar
- Drop down menu
- Utility Classes in bootstrap
- Responsive images and media objects
- Using Bootstrap with JavaScript\
- Building a complete responsive website using Bootstrap

Capstone Project 1 :

- ***"Complete product based website using HTML, CSS, JAVASCRIPT and Bootstrap"***
- ***"Hotel Site and e commerce website"***

Module 8: React (50 hours)

- Understanding React
- Installing node js
- Setup React Project
- CRA (create react app) vs vite
- Folder structure of React App
- How to render in react
- React components
- JSX
- ES Module system
- React Fragments
- Styling in React.js
- Material ui
- Tailwind Css
- Styled components
- Props
- List and Keys
- Conditional Rendering in React
- Event Handling
- State
- Controlled and Uncontrolled components
- React Form Handling
- Form Validation
- Effective state management in React
- Hooks
- useState
- useEffect
- useRef
- useCallback
- useMemo
- useReducer
- Context Api
- Custom Hooks

- Fetching data with Axios
- Routing in React (React-router)
- CSR vs SSR
- Creating Nested Routes
- Dynamic routing
- Lazy Loading
- 404 error
- What is Redux ?
- Setting up Redux in project
- Core Redux vs Redux toolkit
- Actions, reducers and store
- Connecting Redux with React components
- Redux Thunk (Applying middleware)
- synchronous asynchronous
- error handling

Mini Project 5:

- ***"Booking Site"***
- ***"Airbnb clone"***
- ***"Blog Site"***

Module 9: Node Basics (20 hours)

Node JS

- Node
- Installation
- REPL
- NPM
- Command Line Arguments
- Module system in Node(EJS vs CJS)
- Node modules
- Events in Node
- Path Module
- Asynchronous Programming
- File system Module
- OS Module
- Crypto Module
- Zlib Module
- URL Module
- HTTP Module
- Creating a basic HTTP server

- Handling requests and responses
- Serving different type of content
- Handling errors in Node.js
- Debugging Node.js applications
- Child Process in Node

Mini Project 6 :

- ***"Data Fetching from frontend"***
- ***"Connecting React with Express"***

Module 10: Express (20 hours)

Express JS

- What is express?
- What is MVC Pattern?
- Creating Server using Express
- Understanding Request and Response
- Creating routes and handling requests
- Request methods (get ,post, put ,delete)
- Servering HTML in Express
- Setting up Templating Engines
- EJS
- Middlewares
- Serving Static Files
- File uploading using Multer
- Processing data
- Error handling
- Validation
- Mongoose (Connecting Express with MongoDB)
- Create ,delete, upload ,Read Data from mongoDB
- Create Rest api
- Endpoint
- Route API
- Web stocks
- Socket.io
- Scaling

Module 11 : Authentication(10 hours)

- What is Authentication ?
- Session vs Token based Authentication
- JWT
- Implementing JWT in Project
- Implementing login and registration systems
- Protecting routes with middleware
- Password hashing and encryption (e.g., bcrypt)
- Role-based access control

Mini Project 7 :

- ***"Authentication"***
- ***"Make a user Login System"***

Module 12 : MySQL (10 hours)

- What is data base
- mySql database
- Phpmyadmin
- Localhost
- install mysql
- create database
- create table and columns
- drop, delete and alter table
- Insert, select and upload
- Where, from and group by
- Conditions
- Primary keys and foreign keys
- Joins
- Mysql function
- String functions
- Import and export
- CSV
- User Authentication
- Optimizing SQL queries

Project 8 :

- ***"Manage Data with Database"***

Module 13 : MongoDB (10 hours)

MongoDB

- Mongo db overview
- Sql vs nosql
- RDBMS
- Create Database
- Collection
- Insert find and update
- Delete command
- Operators
- CRUD Operations
- Data Modeling in MongoDB
- Scalling
- Security
- Authentication
- Import export
- Aggregations
- Connect to node js
- Validation
- Data API
- Time series
- Geospatial indexing

Project 9 :

- ***"Save Data From Frontend"***
- ***"Data Collection"***

Capstone Project 2 : Anyone One

- ***"Book what you want"***
- ***"Student Git Project"***
- ***"Portfolio"***
- ***"Responsive Website"***

Module 14: Deployment (10 Hours)

- Understanding the deployment
- Uploading the code to cloud
- Deploying code on AWS

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TRAINING COMPLETION CERTIFICATE

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CERTIFICATE
OF PARTICIPATION

Proudly Presented to

Student Name

X Sem |Department of

from X College has successfully completed 6 Months Internship & Training Program in **Full Stack Web Development** from D /M / Y to D /M / Y at **Upflairs Pvt. Ltd.**

Siddharth Singh
Founder & Director
Upflairs Pvt. Ltd.

MSME
MICRO, SMALL & MEDIUM ENTERPRISES
OUR STRENGTH
Ministry of MSME, Govt. of India

#startupindia

Enrollment No. : UF/XXX/XXXX

Trainer Name
Designation
Upflairs Pvt. Ltd.

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OUR STUDENTS FEEDBACK

I'm very happiest to share my experience about my summer internship at Upflair was an incredible learning experience in the data science domain. The hands-on projects, supportive mentors, and real-world applications made it the perfect opportunity for college students to develop their skills. The team's dedication to helping us grow both technically and professionally truly stood out, making it a highly rewarding experience and it is really good to learn something valuable.

Ritu Saini
PIET, Jaipur

I had a good experience at learning in upflairs. We had regular classes and practical implementation of machine learning and data science projects....
Faculty is best as educator and also teaches everything in beginner friendly manner.
We did a many projects and practically learnt the deployment.

Riya Sharma
Arka Jain University

I recently completed my Python internship at Upflairs. The program was well-structured, offering a solid blend of theoretical knowledge and hands-on projects. The mentors were incredibly supportive, providing guidance whenever needed and encouraging creativity and problem-solving.

Sanjay kumar
Arya College of Engineering

It was a great and amazing experience in upflairs pvt ltd, In my 45 days intership of data science and machine learning & AI I have learnt a lot and it helped in making a strong base for data science and ML & AI .

Manvik
PIET, Jaipur

I Anshika Mundra from Poornima Institute of Engineering and Technology have completed my summer training cum internship of 15 days in Frontend Web Development Course. My experience here was fabulous .The knowledgeable and understanding instructors helped me gain various technical and practical skills. I worked on various projects, applying my knowledge to various tasks.

Anshika Mundra
PIET, Jaipur

Great learning experience at Upflairs . Gained valuable skills, worked on meaningful projects, and enjoyed a supportive team environment. Highly recommend for hands-on experience and professional growth. Minor challenges, but overall positive.

Hotri Shrimali
Amity University, Rajasthan

Hello everyone, My name is Antish kumar from career point university. I had enrolled in 2 months Data sciences internship + traning program. The staff is so friendly, helpful to students and resolve all doubts. Our sir Ranjit singh assist me or whole class very well.

Antish Kumar
Career Point University, Kota

It was a really great experience in learning about domain and all the concept of Full stack web development. Environment and level of teaching skills were really up to the mark. Overall, it was a really good experience.

Ajit Prajapat
PIET, Jaipur

I completed my 45-days internship with Er. Ranjit Singh in Upflairs in ye domain of data science with AI&Ml... Here I got a Vry gud xperience with full practical implementation in AI&ML .... The faculty taught all the concepts very well...including the coding part ..... I did multiple projects under his guidance and also developed my final project in the same domain .

Akash Mishra
Arka Jain University

GLANCE

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